

[MS-TPXS]:

Telemetry Protocol XML Schema

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1 Introduction

The Telemetry Protocol XML Schema defines the message structure used by the Software Quality Metrics (SQM) Client to Service Protocol, specified in [\[MS-SQMCS2\]](#). The schema is used to send software instrumentation metrics from a client to the SQM service and for the client to download client-specific control data.

Sections 1.7 and 2 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

Hypertext Transfer Protocol (HTTP): An application-level protocol for distributed, collaborative, hypermedia information systems (text, graphic images, sound, video, and other multimedia files) on the World Wide Web.

Hypertext Transfer Protocol Secure (HTTPS): An extension of HTTP that securely encrypts and decrypts web page requests. In some older protocols, "Hypertext Transfer Protocol over Secure Sockets Layer" is still used (Secure Sockets Layer has been deprecated). For more information, see [\[SSL3\]](#) and [\[RFC5246\]](#).

Secure Sockets Layer (SSL): A security protocol that supports confidentiality and integrity of messages in client and server applications that communicate over open networks. SSL supports server and, optionally, client authentication using X.509 certificates [\[X509\]](#) and [\[RFC5280\]](#). SSL is superseded by Transport Layer Security (TLS). TLS version 1.0 is based on SSL version 3.0 [\[SSL3\]](#).

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

[RFC2616] Fielding, R., Gettys, J., Mogul, J., et al., "Hypertext Transfer Protocol -- HTTP/1.1", RFC 2616, June 1999, <https://www.rfc-editor.org/info/rfc2616>

1.2.2 Informative References

[MS-SQMCS2] Microsoft Corporation, "[Software Quality Metrics \(SQM\) Client-to-Service Version 2 Protocol](#)".

1.3 Overview

This XML data structure specifies a client-to-server request message and a server-to-client response message. The client request message contains one or more commands specifying the work the client is requesting of the server. The server response message contains one response to each request (command) from the client.

1.4 Relationship to Protocols and Other Structures

The Telemetry Protocol XML Schema is used by the SQM Client-to-Service Protocol, as specified in [\[MS-SQMCS2\]](#), to transmit request and response messages between a client and the SQM service.

The request and response data is transmitted over **Hypertext Transfer Protocol (HTTP)** and **Hypertext Transfer Protocol over Secure Sockets Layer (HTTPS)** as specified in [\[RFC2616\]](#).

1.5 Applicability Statement

This structure is defined to support clients that are enabled to request and send telemetry data to a telemetry server. The structure describes the client sender and the data being sent so the server can evaluate each request and either accept or reject the request based on the data contained within the message.

1.6 Versioning and Localization

This specification documents Version 2 of the Telemetry Protocol XML Schema. The XML schema does not support localization.

1.7 Vendor-Extensible Fields

There are no vendor-extensible fields. It is possible for vendors to add key-value arguments to **arg** elements (see the schemas in section [5](#)). However, these key-value pairs are not interpreted unless they are used by the vendor or identified by specific contract between the client and server.

2 Structures

The telemetry data structure describes a telemetry client request and server response.

2.1 Telemetry Request Message

The telemetry request message describes the client request from the server. Each element is described below. The complete schema is specified in the section [5.1](#).

2.1.1 req

The **req** (request) element is the topmost element of a client-to-server request message. It contains an associated version number that is used to differentiate schema changes in the telemetry request message format. The version number of this schema is 2.

```
<xs:element name="req">
  <xs:complexType>
    <xs:attribute name="ver" type="xs:unsignedInt" use="required" />
  </xs:complexType>
</xs:element>
```

2.1.1.1 tlm

The **tlm** (telemetry) element is the namespace of the requested service. The **tlm** element is a child element of [req](#).

```
<xs:element name="tlm">
  <xs:complexType>
  </xs:complexType>
</xs:element>
```

2.1.1.1.1 src

The **src** (source) element is a child element of [tlm](#). The **src** element and the child elements **desc** (description) and **mach** (machine) describe the client that is making the request.

```
<xs:element name="src" minOccurs="1" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="desc" minOccurs="1" maxOccurs="1">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="mach" minOccurs="1" maxOccurs="1">
              <xs:complexType>
                <xs:sequence>
                </xs:sequence>
              </xs:complexType>
            </xs:element>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

2.1.1.1.1.1 os

The **os** is a child element of the **mach** element. It describes the client operating system with a set of key-value pair descriptive attributes.

```
<xs:element name="os" minOccurs="1" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="arg" minOccurs="0" maxOccurs="unbounded" >
        <xs:complexType>
          <xs:attribute name="nm" type="xs:string" use="required" />
          <xs:attribute name="val" type="xs:string" use="required" />
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

The **os** element has a set of child **arg** elements with name-value (**nm, val**) attribute pairs describing the operating system.

nm: Unique key attribute describing a dimension of the client operating system.

val: Value field of the key-value pair describing a dimension of the operating system.

2.1.1.1.1.2 hw

The **hw** (hardware) element is a child element of the **mach** element. It describes the client machine hardware with a set of key-value pair descriptive attributes.

```
<xs:element name="hw" minOccurs="1" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="arg" minOccurs="0" maxOccurs="unbounded" >
        <xs:complexType>
          <xs:attribute name="nm" type="xs:string" use="required" />
          <xs:attribute name="val" type="xs:string" use="required" />
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

The **hw** element has a set of child **arg** elements with name-value (**nm, val**) attribute pairs describing the hardware platform.

nm: Unique key attribute describing a dimension of the client hardware platform.

val: Value field of the key-value pair describing a dimension of the client hardware platform.

2.1.1.1.1.3 ctrl

The **ctrl** (control) element is a child element of the **mach** element. It describes a set of client control values with a set of key-value pair descriptive attributes.

```
<xs:element name="ctrl" minOccurs="1" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="arg" minOccurs="0" maxOccurs="unbounded" >
        <xs:complexType>
          <xs:attribute name="nm" type="xs:string" use="required" />
          <xs:attribute name="val" type="xs:string" use="required" />
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

```

    </xs:complexType>
  </xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>

```

The **ctrl** element has a set of child **arg** elements with name-value (**nm, val**) attribute pairs describing the client control values.

nm: Unique key attribute describing a dimension of the client control values.

val: Value field of the key-value pair describing a dimension of the client control values.

2.1.1.1.2 reqs

The **reqs** (requests) element is the client request section of the message. **reqs** is a child element of [tlm](#).

```

<xs:element name="reqs" minOccurs="1" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
      </xs:sequence>
    </xs:complexType>
  </xs:element>

```

2.1.1.1.2.1 payload

The **payload** element is a child element of **reqs**. It describes the binary data (if any).

```

<xs:element name="payload" minOccurs="0" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
        <xs:complexType>
          <xs:attribute name="nm" type="xs:string" use="required" />
          <xs:attribute name="val" type="xs:string" use="required" />
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>

```

The **payload** element has a set of child **arg** elements with name-value (**nm, val**) attribute pairs describing the type and size of the binary data payload.

nm: Unique key attribute describing a dimension of the payload.

val: Value field of the key-value pair describing a dimension of the payload.

2.1.1.1.2.2 req

The **req** (request) element is a child element of [reqs](#). It describes a client request of the server. The attribute name is **key** and represents a message-wide unique key associated with the **req** element.

```

<xs:element nm="req" minOccurs="1" maxOccurs="unbounded">
  <xs:complexType>
    <xs:attribute nm="key" type="xs:string" use="required"/>
  </xs:complexType>

```



```
</xs:element>
```

2.1.1.1.2.2.1 namespace

The **namespace** element is a child element of **req** and describes the specific client request environment in terms of a set of hierarchical namespaces. The namespace is order-defined as the 5tuple **root.svc.ptr.gp.app. tlm** is the implied value of the **root** component of the tuple.

```
<xs:element name="namespace" minOccurs="1" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
        <xs:complexType>
          <xs:attribute name="nm" type="xs:string" use="required" />
          <xs:attribute name="val" type="xs:string" use="required" />
        </xs:complexType>
      </xs:element>
    </xs:sequence>
    <xs:attribute name="svc" type="xs:string" use="required" />
    <xs:attribute name="ptr" type="xs:string" use="required" />
    <xs:attribute name="gp" type="xs:string" use="required" />
    <xs:attribute name="app" type="xs:string" use="required" />
  </xs:complexType>
</xs:element>
```

The **namespace** element has a set of child **arg** elements with name-value (**nm, val**) attribute pairs that serve as additional namespace descriptors.

nm: Unique key attribute describing a dimension of the namespace.

val: Value field of the key-value pair describing a dimension of the namespace.

The **namespace** element has the following attributes:

svc: An attribute that describes the client service sending the request.

ptr: An attribute that describes the client service partner sending the request. **ptr** differentiates the service namespace (**root.svc.ptr**).

gp: An attribute that describes the client service partner group sending the request. **gp** differentiates the service namespace (**root.svc.ptr.gp**).

app: An attribute that describes the client service partner group application sending the request. **app** further differentiates the group namespace (**root.svc.ptr.gp.app**).

2.1.1.1.2.2.2 ctrl

The **ctrl** (control) element is a child element of **req** and describes the control data values for the request with a set of name-value pairs. **ctrl** arguments are specific to the **req** element whereas the machine-level **ctrl** (see section [2.1.1.1.1.3](#)) arguments are client machine-wide.

```
<xs:element name="ctrl" minOccurs="0" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
        <xs:complexType>
          <xs:attribute name="nm" type="xs:string" use="required" />
          <xs:attribute name="val" type="xs:string" use="required" />
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

```

    </xs:sequence>
  </xs:complexType>
</xs:element>

```

The **ctrl** element has a set of child **arg** elements with name-value (**nm, val**) attribute pairs describing the client request control values.

nm: Unique key attribute describing a dimension of the client request control values.

val: Value field of the key-value pair describing a dimension of the client request control values.

2.1.1.1.2.2.3 contents

The **contents** element is a child element of **req** and describes the payload content (if any).

```

<xs:element name="contents" minOccurs="0" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
        <xs:complexType>
          <xs:attribute name="nm" type="xs:string" use="required" />
          <xs:attribute name="val" type="xs:string" use="required" />
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>

```

The **contents** element has a set of child **arg** elements with name-value (**nm, val**) attribute pairs describing the contents of the client request payload.

nm: Name attribute describing a dimension of the payload contents. The name is not required to be unique.

val: Value field of the name-value pair describing a dimension of the payload contents.

2.1.1.1.2.2.4 cmd

The **cmd** (command) element is a child element of the **req** element. It describes the client request work from the server. The **cmd** element has one attribute **nm**, a name value describing the command. The server uses the name to determine what work is requested.

```

<xs:element name="cmd" minOccurs="1" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
        <xs:complexType>
          <xs:attribute name="nm" type="xs:string" use="required" />
          <xs:attribute name="val" type="xs:string" use="required" />
        </xs:complexType>
      </xs:element>
    </xs:sequence>
    <xs:attribute name="nm" type="xs:string" use="required" />
  </xs:complexType>
</xs:element>

```

The **cmd** element has a set of child **arg** elements with name-value (**nm, val**) attribute pairs describing the command arguments.

nm: Unique key attribute describing an argument of the command.

val: Value field of the key-value pair describing an argument of the command.

2.2 Telemetry Response Message

The telemetry response message describes the server response to a client telemetry request message. Each element is described in the following sections. The complete schema is specified in section [5.2](#).

2.2.1 resp

The **resp** (response) element is the topmost element of a server-to-client response message. There is an associated version number. The version number is used to differentiate schema changes in the telemetry response message format. The version number of this schema is 2.

```
<xs:element name="resp">
  <xs:complexType>
    <xs:attribute name="ver" type="xs:unsignedInt" use="required" />
  </xs:complexType>
</xs:element>
```

2.2.1.1 tlm

The **tlm** (telemetry) element is a child element of the **resp** element. It is the namespace of the service.

```
<xs:element name="tlm">
  <xs:complexType>
  </xs:complexType>
</xs:element>
```

2.2.1.1.1 resps

The **resps** (responses) element is a child element of the **tlm** element. It contains the server responses to the client requests.

```
<xs:element name="resps" minOccurs="1" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

2.2.1.1.1.1 resp

The **resp** (response) element describes a server response to a client request. **resp** is a child element of [resps](#). **key** is an attribute and maps to the client **req** key in the client telemetry request message. There is a one-to-one mapping of the telemetry response message **resp** element to the telemetry request message **req** element (see section [2.1.1.1.2.2](#)).

```
<xs:element nm="resp" minOccurs="1" maxOccurs="unbounded">
  <xs:complexType>
    <xs:attribute nm="key" type="xs:string" use="required" />
  </xs:complexType>
</xs:element>
```

2.2.1.1.1.1 namespace

The **namespace** element is echoed back from the telemetry request message **req** (section [2.1.1](#)) unaltered. **namespace** is a child element of **resp**. See the telemetry request message **namespace** element as specified in section [2.1.1.1.2.2.1](#).

```
<xs:element name="namespace" minOccurs="1" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
        <xs:complexType>
          <xs:attribute name="nm" type="xs:string" use="required" />
          <xs:attribute name="val" type="xs:string" use="required" />
        </xs:complexType>
      </xs:element>
    </xs:sequence>
    <xs:attribute name="svc" type="xs:string" use="required" />
    <xs:attribute name="ptr" type="xs:string" use="required" />
    <xs:attribute name="gp" type="xs:string" use="required" />
    <xs:attribute name="app" type="xs:string" use="required" />
  </xs:complexType>
</xs:element>
```

2.2.1.1.1.2 cmd

The **cmd** (command) element is a child element of the **resp** element and describes the server command response to the client. The **cmd** element has one attribute, an **nm** (name) attribute describing the command.

```
<xs:element name="cmd" minOccurs="1" maxOccurs="unbounded">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
        <xs:complexType>
          <xs:attribute name="nm" type="xs:string" use="required" />
          <xs:attribute name="val" type="xs:string" use="required" />
        </xs:complexType>
      </xs:element>
    </xs:sequence>
    <xs:attribute name="nm" type="xs:string" use="required" />
  </xs:complexType>
</xs:element>
```

The **cmd** element has a set of child **arg** elements with name-value (**nm, val**) attribute pairs describing the command arguments.

nm: Name attribute describing an argument of the command. The name is not required to be unique.

val: Value field of the key-value pair describing an argument of the command.

3 Structure Examples

3.1 Client-to-Server Request

This section contains an example of an SQM client-to-server request to upload SQM data as indicated by the **requpload** command. There are two requests (key 1,2), each with a separate **requpload** command.

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<req ver="2">
  <tlm>
    <src>
      <desc>
        <mach>
          <os>
            <arg nm="vermaj" val="6" />
            <arg nm="vermin" val="2" />
            <arg nm="verbld" val="8044" />
            <arg nm="verqfe" val="0" />
            <arg nm="versp" val="0" />
            <arg nm="arch" val="0" />
            <arg nm="lcid" val="1033" />
            <arg nm="geoid" val="244" />
          </os>
          <hw>
            <arg nm="form" val="5" />
            <arg nm="arch" val="9" />
            <arg nm="sysmfg" val="Dell Inc." />
            <arg nm="syspro" val="Precision WorkStation 380" />
            <arg nm="bv" val="A07" />
            <arg nm="ram" val="2048" />
          </hw>
          <ctrl>
            <arg nm="tm" val="129552509093248060" />
            <arg nm="mid" val="{1BC55FD8-3C15-4183-9E34-D8DCCE90535E}" />
          </ctrl>
        </mach>
      </desc>
    </src>
    <reqs>
      <req key="1">
        <namespace svc="sqm" ptr="windows" gp="winsqm8" app="13238528"></namespace>
        <ctrl>
          <arg nm="uid" val="{528EC640-26D0-48CC-9609-E2E44D9194C1}" />
        </ctrl>
        <cmd nm="requpload"></cmd>
      </req>
      <req key="2">
        <namespace svc="sqm" ptr="windows" gp="winsqm8" app="1"></namespace>
        <ctrl>
          <arg nm="uid" val="{528EC640-26D0-48CC-9609-E2E44D9194C1}" />
        </ctrl>
        <cmd nm="requpload"></cmd>
      </req>
    </reqs>
  </tlm>
</req>
```

3.2 Server-to-Client Response

This section contains an example of an SQM server-to-client response. There are two responses that map to the requests in section [3.1](#).

```
<?xml version="1.0" encoding="utf-8"?>
<resp ver="2">
  <tlm>
    <resps>
      <resp key="1">
        <namespace svc="sqm" ptr="windows" gp="winsqm8" app="13238528">
        </namespace>
        <cmd nm="throttle">
          <arg nm="period" val="30" />
          <arg nm="namespace" val=" app" />
        </cmd>
      </resp>
      <resp key="2">
        <namespace svc="sqm" ptr="windows" gp="winsqm8" app="1">
        </namespace>
        <cmd nm="approved">
          <arg nm="token" val="1.5c719a32ffe543c0.1cb128940d0e3a7" />
          <arg nm="tokenexp" val="129561149070780000" />
        </cmd>
      </resp>
    </resps>
  </tlm>
</resp>
```

4 Security

4.1 Security Considerations for Implementers

The Telemetry Protocol XML Schema relies on the implementor to secure the message over a network by using a secure transport such as **SSL**. Telemetry Protocol XML Schema does not have built-in support for security or authentication.

4.2 Index Of Security Fields

None.

5 Appendix A: Full XML Schema

5.1 Telemetry Request Message - Full Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema attributeFormDefault="unqualified" elementFormDefault="qualified"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="req">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="tlm">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="src" minOccurs="1" maxOccurs="1">
                <xs:complexType>
                  <xs:sequence>
                    <xs:element name="desc" minOccurs="1" maxOccurs="1">
                      <xs:complexType>
                        <xs:sequence>
                          <xs:element name="mach" minOccurs="1" maxOccurs="1">
                            <xs:complexType>
                              <xs:sequence>
                                <xs:element name="os" minOccurs="1" maxOccurs="1">
                                  <xs:complexType>
                                    <xs:sequence>
                                      <xs:element name="arg" minOccurs="0" maxOccurs="unbounded" >
                                        <xs:complexType>
                                          <xs:attribute name="nm" type="xs:string" use="required" />
                                          <xs:attribute name="val" type="xs:string" use="required" />
                                        </xs:complexType>
                                      </xs:element>
                                    </xs:sequence>
                                  </xs:complexType>
                                </xs:element>
                              </xs:sequence>
                            </xs:complexType>
                          </xs:element>
                        <xs:element name="hw" minOccurs="1" maxOccurs="1">
                          <xs:complexType>
                            <xs:sequence>
                              <xs:element name="arg" minOccurs="0" maxOccurs="unbounded" >
                                <xs:complexType>
                                  <xs:attribute name="nm" type="xs:string" use="required" />
                                  <xs:attribute name="val" type="xs:string" use="required" />
                                </xs:complexType>
                              </xs:element>
                            </xs:sequence>
                          </xs:complexType>
                        </xs:element>
                      <xs:element name="ctrl" minOccurs="1" maxOccurs="1">
                        <xs:complexType>
                          <xs:sequence>
                            <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
                              <xs:complexType>
                                <xs:attribute name="nm" type="xs:string" use="required" />
                                <xs:attribute name="val" type="xs:string" use="required" />
                              </xs:complexType>
                            </xs:element>
                          </xs:sequence>
                        </xs:complexType>
                      </xs:element>
                    </xs:sequence>
                  </xs:complexType>
                </xs:element>
              </xs:sequence>
            </xs:complexType>
          </xs:element>
        </xs:sequence>
      </xs:complexType>
    </xs:element>
  </xs:sequence>
</xs:schema>
```



```

</xs:element>
<xs:element name="regs" minOccurs="1" maxOccurs="1">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="payload" minOccurs="0" maxOccurs="1">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
              <xs:complexType>
                <xs:attribute name="nm" type="xs:string" use="required" />
                <xs:attribute name="val" type="xs:string" use="required" />
              </xs:complexType>
            </xs:element>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
      <xs:element name="req" minOccurs="1" maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="namespace" minOccurs="1" maxOccurs="1">
              <xs:complexType>
                <xs:sequence>
                  <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
                    <xs:complexType>
                      <xs:attribute name="nm" type="xs:string" use="required" />
                      <xs:attribute name="val" type="xs:string" use="required" />
                    </xs:complexType>
                  </xs:element>
                </xs:sequence>
                <xs:attribute name="svc" type="xs:string" use="required" />
                <xs:attribute name="ptr" type="xs:string" use="required" />
                <xs:attribute name="gp" type="xs:string" use="required" />
                <xs:attribute name="app" type="xs:string" use="required" />
              </xs:complexType>
            </xs:element>
            <xs:element name="ctrl" minOccurs="0" maxOccurs="1">
              <xs:complexType>
                <xs:sequence>
                  <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
                    <xs:complexType>
                      <xs:attribute name="nm" type="xs:string" use="required" />
                      <xs:attribute name="val" type="xs:string" use="required" />
                    </xs:complexType>
                  </xs:element>
                </xs:sequence>
              </xs:complexType>
            </xs:element>
            <xs:element name="contents" minOccurs="0" maxOccurs="1">
              <xs:complexType>
                <xs:sequence>
                  <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
                    <xs:complexType>
                      <xs:attribute name="nm" type="xs:string" use="required" />
                      <xs:attribute name="val" type="xs:string" use="required" />
                    </xs:complexType>
                  </xs:element>
                </xs:sequence>
              </xs:complexType>
            </xs:element>
            <xs:element name="cmd" minOccurs="1" maxOccurs="1">
              <xs:complexType>
                <xs:sequence>
                  <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
                    <xs:complexType>
                      <xs:attribute name="nm" type="xs:string" use="required" />
                      <xs:attribute name="val" type="xs:string" use="required" />
                    </xs:complexType>
                  </xs:element>
                </xs:sequence>
              </xs:complexType>
            </xs:element>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>

```

```

        <xs:attribute name="nm" type="xs:string" use="required" />
      </xs:complexType>
    </xs:element>
  </xs:sequence>
  <xs:attribute name="key" type="xs:string" use="required" />
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:sequence>
<xs:attribute name="ver" type="xs:unsignedInt" use="required" />
</xs:complexType>
</xs:element>
</xs:schema>

```

5.2 Telemetry Response Message - Full Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema attributeFormDefault="unqualified" elementFormDefault="qualified"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="resp">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="tlm">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="resps" minOccurs="1" maxOccurs="1">
                <xs:complexType>
                  <xs:sequence>
                    <xs:element name="resp" minOccurs="1" maxOccurs="unbounded">
                      <xs:complexType>
                        <xs:sequence>
                          <xs:element name="namespace" minOccurs="1" maxOccurs="1">
                            <xs:complexType>
                              <xs:sequence>
                                <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
                                  <xs:complexType>
                                    <xs:attribute name="nm" type="xs:string" use="required" />
                                    <xs:attribute name="val" type="xs:string" use="required" />
                                  </xs:complexType>
                                </xs:element>
                              </xs:sequence>
                            </xs:complexType>
                          <xs:attribute name="svc" type="xs:string" use="required" />
                          <xs:attribute name="ptr" type="xs:string" use="required" />
                          <xs:attribute name="gp" type="xs:string" use="required" />
                          <xs:attribute name="app" type="xs:string" use="required" />
                        </xs:complexType>
                      </xs:element>
                    <xs:element name="cmd" minOccurs="1" maxOccurs="unbounded">
                      <xs:complexType>
                        <xs:sequence>
                          <xs:element name="arg" minOccurs="0" maxOccurs="unbounded">
                            <xs:complexType>
                              <xs:attribute name="nm" type="xs:string" use="required" />
                              <xs:attribute name="val" type="xs:string" use="required" />
                            </xs:complexType>
                          </xs:element>
                        </xs:sequence>
                      </xs:complexType>
                    </xs:element>
                    <xs:attribute name="nm" type="xs:string" use="required" />
                  </xs:complexType>
                </xs:element>
              </xs:sequence>
            </xs:complexType>
          </xs:element>
          <xs:attribute name="key" type="xs:string" use="required" />
        </xs:sequence>
      </xs:complexType>
    </xs:element>
  </xs:schema>

```

```
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:sequence>
<xs:attribute name="ver" type="xs:unsignedInt" use="required" />
</xs:complexType>
</xs:element>
</xs:schema>
```

6 Appendix B: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

- Windows 8 operating system
- Windows Server 2012 operating system
- Windows 8.1 operating system
- Windows Server 2012 R2 operating system
- Windows 10 operating system
- Windows Server 2016 operating system
- Windows Server operating system
- Windows Server 2019 operating system
- Windows Server 2022 operating system
- Windows 11 operating system
- Windows Server 2025 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

7 Change Tracking

This section identifies changes that were made to this document since the last release. Changes are classified as Major, Minor, or None.

The revision class **Major** means that the technical content in the document was significantly revised. Major changes affect protocol interoperability or implementation. Examples of major changes are:

- A document revision that incorporates changes to interoperability requirements.
- A document revision that captures changes to protocol functionality.

The revision class **Minor** means that the meaning of the technical content was clarified. Minor changes do not affect protocol interoperability or implementation. Examples of minor changes are updates to clarify ambiguity at the sentence, paragraph, or table level.

The revision class **None** means that no new technical changes were introduced. Minor editorial and formatting changes may have been made, but the relevant technical content is identical to the last released version.

The changes made to this document are listed in the following table. For more information, please contact dochelp@microsoft.com.

Section	Description	Revision class
6 Appendix B: Product Behavior	Added Windows Server 2025 to the list of applicable products.	Major

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